AnchorMTM: Spatially Anchored Merge Tree Maps for Time-Varying Scalar Fields

Category: Research

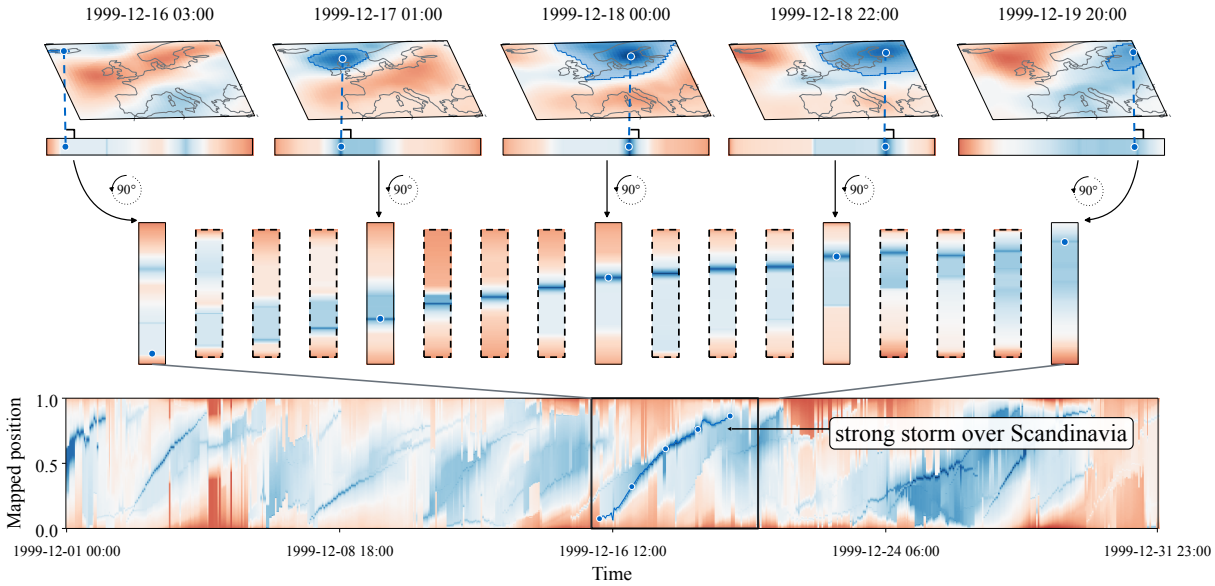


Figure 1: Storms example linking spatial feature supports to a temporal map ($\varepsilon = 5\%$). Top: five snapshots of one matched extremum and their scalar profiles; blue outlines mark its leaf-arc supports, and dashed guides link the extrema to their displayed profile anchors. Middle: the profiles rotated by 90°; dashed borders distinguish sampled intermediate frames. Bottom: the full 744-frame map, with the selected storm-related region outlined and the matched feature's displayed positions marked in blue. The vertical map axis is the normalized reference coordinate, not longitude.

Abstract

Temporal merge tree maps show how the topology of a scalar field evolves in one image, but their display axis has no fixed spatial meaning, so a feature's position cannot be read as its location. We anchor this axis to a spatial direction, estimated from feature motion or supplied by the analyst. Anchoring collides with the hierarchy: the merge tree keeps subtrees contiguous, while their projections may interleave. We prove three lower bounds on the displacement this conflict forces under exact topology, finite feature widths and a bounded merge error. The bounds hold for every admissible layout, so they also measure existing maps. AnchorMTM builds the map with one control, a merge-error budget ε, through a leaf-order search with optimal merge barriers and a capped quadratic program. On seven planar and volumetric sequences (917 frames), Exact Mode meets the width-aware bound on 606 of 614 planar frames, so no exact-topology order has a smaller worst displacement there, while 31–95% of the order error of TMTM and ST-MTM is not forced by the tree. Raising ε trades merge-level error for position, and every rendered map stays within its budget.

Index Terms: Time-varying scalar fields, merge tree maps, spatial anchoring, topology preservation, constrained layout.

1 Introduction

An analyst who follows storms across an ocean or convection across the tropics reads a temporal overview for three things at once: which features exist and how they nest, where they are, and how they move. Static overviews give one display axis to time and compress space into the other [1, 31, 34]; what that second axis means decides which of these questions the overview can answer. Merge tree maps answer the first one well. Temporal Merge Tree Maps (TMTM) draw the merge tree of each frame as a 1D scalar profile and align the profiles along time, so minima show extrema and the barriers between them show merge levels [22]. Spatiotemporal Merge Tree Maps (ST-MTM) add relative spatial distances and temporal position constraints to the layout [9].

Neither map gives its display axis a fixed spatial meaning. TMTM orders leaves for the tree and for temporal alignment; ST-MTM preserves pairwise distances, which fix relative positions but no origin or direction. A feature's position therefore cannot be read as its longitude or its distance along a flow, and a feature that moves east need not move on the map. Hovmöller diagrams take the opposite side: they keep one physical coordinate exactly and average the field over the others, so nearby minima blend and the hierarchy disappears [20]. Figure 1 shows what we want from both: a storm followed in space and in the map, on an axis that runs along a fixed spatial direction.

Anchoring the axis means projecting every extremum onto one direction and placing it at its projection. The merge tree resists. An exact profile keeps every subtree in one contiguous block, but the projections of sibling subtrees can interleave. In Figure 2, four minima project in the order A C B D and split both sibling pairs, so some features must leave their references or some merge levels must change. Whether a spatial axis is worth having depends on how much displacement the hierarchy forces and how much merge error buys it back.

We answer both questions. Three lower bounds give the displacement that exact topology, finite feature widths and a bounded merge error force on every admissible layout; they depend on the tree, the references and the widths, and on no solver. AnchorMTM

builds a map that approaches them. It estimates one reference direction from matched extremum motion or takes one from the analyst, searches for a leaf order within a merge-error budget ε, sets optimal merge barriers and places anchors with a capped quadratic program. Exact Mode ($\varepsilon = 0$) keeps the merge tree; Budgeted Mode ($\varepsilon > 0$) trades a declared merge-level error for position. On seven planar and volumetric sequences with 917 frames, the leaf order of Exact Mode meets the width-aware bound on 606 of 614 planar frames and is therefore optimal there. The exact-topology bound shows that 31–95% of the order error of TMTM and ST-MTM is not forced by the tree; on planar data the tree forces at most 5.8% of the axis on average. A spatial axis costs the hierarchy far less than existing maps suggest.

Our contributions are:

- a formulation of spatially anchored merge tree maps as hierarchy-constrained interval layouts with a fixed reference and a merge-level allowance, with three lower bounds on the displacement that exact topology, finite widths and a bounded merge error force on any such layout (Section 4.2);
- AnchorMTM, a construction with one control, the merge-error budget ε, built from topology-budgeted leaf ordering, optimal merge barriers and capped anchor placement (Sections 4.3 and 4.4);
- an evaluation on seven sequences showing that the bounds are attained in controlled trees and met by the selected order on almost every planar frame, and that most of the order error of TMTM and ST-MTM is not forced by the hierarchy (Section 5).

2 Related Work

2.1 Spatially Referenced Temporal Visualization

Animation keeps the spatial view of each instant, while static displays expose distant times together; which serves better depends on the task [31, 34]. Generalized space–time cubes describe the transformations between such views [1].

Hovmöller diagrams retain a selected physical coordinate and summarize the other dimensions [20]. Geographic orderings and space-filling curves instead traverse the domain [8, 16]. Dynamic Volume Lines and data-driven curves extend spatial traversal to volumetric comparison and scalar-aware ordering [37, 39]. A traversal keeps spatial neighborhoods, but its coordinate is not a fixed projection and its adjacency does not enforce the merge hierarchy.

1D projections organize spatiotemporal phenomena [14], and MotionRugs expose collective behavior of moving entities [5]. More generally, an embedding trades original-domain relationships against its display organization [28]. AnchorMTM assigns fixed projection targets to topological features and bounds their displacement under hierarchical and interval constraints.

2.2 Topology-Based Temporal Maps and Geometry-Aware Layouts

Topology-based visualization organizes scalar structures through connectivity [17]. Landscapes and landscape profiles express hierarchies through terrain or 1D scalar representations [29, 35], with geometry-preserving and temporally coherent extensions [3, 18].

TMTM emphasizes intact topological features and temporal alignment in static merge-tree maps [22]. ST-MTM incorporates relative spatial geometry and temporal position constraints [9]. Neither ties the display coordinate to a physical direction. AnchorMTM adds that reference, bounds what it costs, and reuses their profile construction and scalar filling.

Hierarchy-constrained leaf ordering traditionally minimizes adjacent dissimilarity [2]; target-order reordering studies related tree-consistent permutations [6]. Separation-constrained layouts control geometric overlap and displacement [11, 24]. AnchorMTM couples these layout concerns to fixed spatial targets. Its exact-topology point-order bound has a closed form, whereas finite-width candidate ordering remains heuristic.

Merge-tree comparison depends on the descriptors and correspondences used [38]. Labelled cophenetic/interleaving comparisons support fixed-identity merge-level accounting [15, 27], and ordered merge trees connect scalar curves to monotone interleaving distances [4]. The optimal-barrier subproblem relates to minimax ultrametric fitting [13], with the additional restriction that one linear order induces every displayed merge.

3 Background

3.1 Augmented Merge Trees and Feature Supports

For a sampled 2D or 3D field $f_t : D \to \mathbb{R}$, the merge tree T_t records sublevel-set connectivity: minima create components and saddles join them as the threshold increases. Negating the field gives the split-tree convention. Critical points are supernodes connected by superarcs; augmentation associates regular vertices with these arcs [9, 22, 32]. Extraction and simplification precede layout.

A leaf represents an extremum at spatial location p_i^t. Its feature support S_i^t consists of that extremum and the regular samples on its incident leaf arc, excluding the parent critical sample. It is not the full component or the complete subtree above that arc. Supports are sets of sample IDs that denote the same spatial sample in every frame, as on a fixed grid. Internal arcs retain further regular samples for scalar filling. Write $\mathscr{D}_t(v)$ for the descendant leaf labels of a node v, and $h_t(v)$ for its scalar level. The lowest common ancestor determines the merge-height matrix:

$$V_{ij}^t = h_t(\mathrm{lca}(i,j)), \quad i \neq j; \qquad V_{ii}^t = f_t(p_i^t). \tag{1}$$

Equal-level merge events form one multifurcation; arbitrary binary refinements do not add strict constraints. All subsequent guarantees refer to the supplied, possibly simplified tree.

3.2 One-Dimensional Merge-Tree Profiles

A 1D merge-tree profile represents extrema at anchors in a selected leaf order. Regular vertices on leaf arcs supply scalar context within leaf intervals; internal arcs and paths to the LCA supply the intervening gaps [9, 22]. In join convention, extrema become profile minima, and the largest profile value between two anchors gives their displayed merge level. Exact preservation requires each strict subtree to occupy a contiguous leaf block. Arranging these profiles along time yields a static temporal map.

4 AnchorMTM

Section 4.1 states the layout model, the admissible maps and the measures. Section 4.2 bounds the displacement that every admissible map must accept. AnchorMTM then fixes a spatial reference (Section 4.3) and builds, with one control, a map that approaches these bounds (Section 4.4). Analysis uses join convention; split trees are negated and their sign is restored after reconstruction.

4.1 Layout Model and Fidelity Measures

A merge tree map chooses, per frame, a leaf order, extremum anchors, and a scalar profile:

$$M_t = (\pi_t, x_t, \psi_t), \qquad \psi_t : [0,L] \to \mathbb{R}. \tag{2}$$

Leaf i has a reference position $q_i \in [0,L]$ on the display axis (Section 4.3 constructs it) and a width $W_i^t = s m_i^t$ from its positive support measure with a fixed scale s, without per-frame normalization. The interval-layout set $\mathscr{L}(\pi)$ requires intervals of widths W_i centered at z_i to lie inside the canvas, remain disjoint with gap at least $g \geq 0$, and contain anchors satisfying $|x_i - z_i| \leq \rho W_i/2$, $0 \leq \rho \leq 1$.

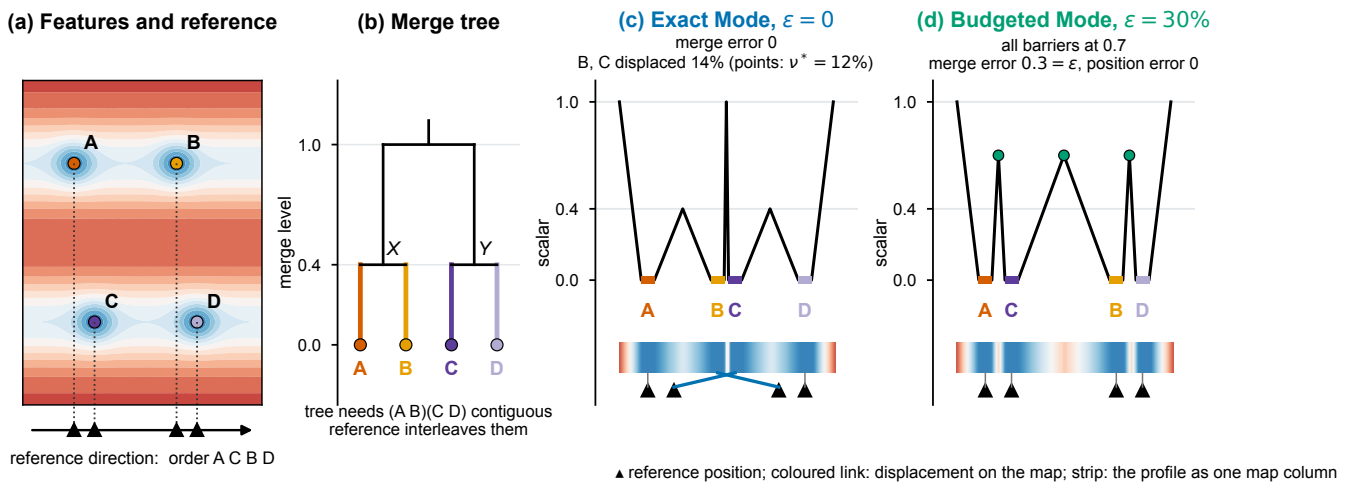


Figure 2: Why anchoring conflicts with the hierarchy, and what the allowance buys, on the four-leaf example of Section 5.2. (a) Projected onto the reference, the minima are ordered A C B D; (b) the merge tree requires the sibling blocks (A B) and (C D). (c) Exact Mode keeps the blocks contiguous and displaces B and C by 14% (12% for points, Theorem 1). (d) With an allowance of half the parent gap, $\varepsilon = 30\%$, the reference order is admissible: all three barriers are set to 0.7, so every displayed merge level is off by exactly $0.3 = \varepsilon$, and no feature is displaced. Each strip below a profile is the corresponding column of the temporal map.

Widths are never rescaled; a frame whose widths and gaps exceed the canvas is reported.

Actual displacement is $|x_i - q_i|$, normalized by L when reported. To distinguish order conflict from finite-width placement, define

$$\mathrm{NROE}(\pi) = \frac{1}{2L} \max_{j \le k} [q_{\pi(j)} - q_{\pi(k)}]^+. \tag{3}$$

This is the smallest worst displacement of a nondecreasing point placement, divided by L; it ignores widths. For finite intervals, the exact fixed-order cost is

$$\tau(\pi) = \min_{(x,z) \in \mathscr{L}(\pi)} \|x - q\|_\infty. \tag{4}$$

The minimum is attained whenever the layout set is nonempty. It is evaluated exactly as a minimax linear program.

For distinct labels, let V^{ψ}_{ij} be the maximum profile value between their anchors, with $V^{\psi}_{ii} = \psi(x_i)$. A map admits merge-level allowance $\eta \ge 0$ if

$$d_{\mathrm{top}}(\psi_t, T_t) = \max_{i,j} |V^{\psi_t}_{ij} - V^{t}_{ij}| \le \eta. \tag{5}$$

AnchorMTM sets $\eta = (\varepsilon_0 + \varepsilon)\Delta_F$, where $\Delta_F > 0$ is the sequence-wide scalar range, ε_0 is a fixed numerical allowance, and $\varepsilon \ge 0$ is the sole primary user control. The default $\varepsilon_0 = 10^{-7}$ only merges saddles tied to within $\varepsilon_0 \Delta_F$. NMLE reports the achieved normalized error $d_{\mathrm{top}}/\Delta_F$. Strict exact topology is $\eta = 0$: zero labelled error on the declared tree.

For canonical cophenetic merge-tree representations whose fixed labels cover every leaf and whose induced matrices include the diagonal, d_{top} equals the unnormalized labelled interleaving distance [15, 27]. Under these conditions, $d_B \le d_I \le d_{\mathrm{top}}$ [26].

A map is *admissible* for allowance η if its intervals form a layout in $\mathscr{L}(\pi_t)$ with anchors x_t, its profile shows every extremum at its value, $\psi_t(x^t_i) = V^t_{ii}$, and $d_{\mathrm{top}}(\psi_t, T_t) \le \eta$. Let $\Pi_\eta(T_t)$ contain the orders that admit such a profile; Proposition 1 gives the exact test $\delta^*_t(\pi) \le \eta$. For $\eta = 0$, every strict subtree occupies a contiguous block. The bounds of Section 4.2 concern admissible maps only, whatever their objective or solver; AnchorMTM chooses among them by a fixed reference, a position cap and the objective of Eq. 19.

4.2 Certified Position–Topology Bounds

How far must an admissible map leave its references? The first bound answers this for a tree-compatible order of points, the second adds interval packing, and the third asks which order restrictions survive a merge-error allowance. Each bound depends only on the tree, the references and the widths, so it applies to any layout method; full proofs are in Supplement S2–S5. We suppress t within a frame, use $[v]^+ = \max(v, 0)$, and take empty maxima as zero.

4.2.1 Exact-Topology Reference-Order Bound

For a reduced tree with strictly increasing proper-ancestor levels, exact topology makes every subtree contiguous. Let $[m_X, M_X]$ be the reference hull of the leaves in subtree X.

Theorem 1 (Exact-Topology Position Bound)**.** *Every zero-error profile has* $\mathrm{NROE}(\pi) \ge \nu^*(T, q)$*, where*

$$\nu^*(T, q) = \frac{1}{2L} \max_{v} \max_{\substack{X, Y \in \mathrm{ch}(v) \\ X \ne Y}} [\min(M_X - m_Y, M_Y - m_X)]^+. \tag{6}$$

Recursive child ordering by reference-hull midpoint attains this point-model order bound in $O(n \log n)$ *time.*

Proof sketch. Sibling blocks must choose one orientation. Their two possible inversion costs are $[M_X - m_Y]^+/2$ and $[M_Y - m_X]^+/2$. Midpoint order chooses the smaller; every inverted leaf pair is separated by sibling blocks at its LCA. □

Finite intervals add further constraints, considered next.

4.2.2 Width-Aware Position Bound

Finite widths, minimum gaps, anchor slack and canvas capacity add packing costs to the point-order obstruction. For a nonempty leaf set S, let $H_i = \rho W_i/2$, $\ell_i = q_i - W_i/2 - H_i$, $h_i = q_i + W_i/2 + H_i$, and

$s_i = W_i + g$. Define two packing summaries:

$$A(S) = \max_{a \in S} \left(\ell_a + \sum_{\substack{i \in S \\ \ell_i \geq \ell_a}} s_i \right), \qquad B(S) = \max_{b \in S} \left(\sum_{\substack{i \in S \\ h_i \leq h_b}} s_i - h_b \right). \tag{7}$$

The first uses upper reference-endpoint sets and the second lower sets; sorting and cumulative sums evaluate them. For disjoint leaf sets X, Y and the full leaf set $\mathscr{F}$, write

$$\omega(X,Y) = \tfrac{1}{2}\min\{A(X)+B(Y), A(Y)+B(X)\} - \tfrac{g}{2}, \qquad E_\partial = \max\{0, B(\mathscr{F}) - g, A(\mathscr{F}) - g - L\}. \tag{8}$$

Here ω takes the cheaper block orientation and E_∂ accounts for both canvas ends. Let $\mathscr{P}_0$ contain every pair of distinct sibling descendant-leaf sets. The certificate is

$$\mathscr{B}_w(T,q,W) = \max\{E_\partial, \max_{(X,Y)\in\mathscr{P}_0} \omega(X,Y)\}. \tag{9}$$

The dependence on g, ρ and L is through Eqs. 7–8; it is suppressed in the certificate's arguments.

Theorem 2 (Width-Aware Position Bound)**.** *For any feasible exact-topology interval layout,*

$$\|x - q\|_\infty \geq \tau(\pi) \geq \mathscr{B}_w(T,q,W). \tag{10}$$

With zero widths and gaps, $\mathscr{B}_w = L\nu^*(T,q)$.

Proof sketch. The first and last anchors of selected leaves must enclose their intervals and intervening gaps, up to the allowed anchor slack. Upper and lower sets in Eq. 7 maximize the resulting shortfall against reference separation. Exact topology requires one orientation of every sibling pair; the canvas ends give the two one-sided costs. □

Thus $\mathscr{B}_w$ bounds all exact-topology legal interval orders and need not be attained; $\tau(\pi)$ is the attained minimum for a specified order. These quantities have coordinate units, whereas NROE and ν^* are normalized by L.

4.2.3 Bounded-Topology Position Bound

A merge-level allowance can release some contiguous blocks, but still limits interleaving. If leaf k lies between descendants i, j of node u, interval containment gives $V_{ik}^{\Psi} \leq V_{ij}^{\Psi}$, hence $V_{ik} \leq h(u) + 2\eta$. Define $A_\eta(u)$ as the highest ancestor, including u, with $h(A_\eta(u)) - h(u) \leq 2\eta$. This uses the accumulated ancestor-height difference in scalar units, on the reduced equal-level tree with the fixed labels of Eq. 5.

Let $\mathscr{P}_\eta$ consist of two types of leaf-set pairs: $(\mathscr{D}(u), \{k\})$ for every internal node u and leaf $k \notin \mathscr{D}(A_\eta(u))$; and $(\mathscr{D}(u), \mathscr{D}(w))$ for internal nodes u, w satisfying

$$\mathscr{D}(w) \cap \mathscr{D}(A_\eta(u)) = \varnothing, \qquad \mathscr{D}(u) \cap \mathscr{D}(A_\eta(w)) = \varnothing. \tag{11}$$

For any nonempty leaf set X, use its reference hull $[m_X, M_X]$ and define $\kappa(X,Y) = [\min(M_X - m_Y, M_Y - m_X)]^+/(2L)$. The two certificates are

$$C_\eta(T,q) = \max_{(X,Y)\in\mathscr{P}_\eta} \kappa(X,Y), \qquad C_{w,\eta}(T,q,W) = \max\{E_\partial, \max_{(X,Y)\in\mathscr{P}_\eta} \omega(X,Y)\}. \tag{12}$$

Theorem 3 (Bounded-Topology Position Bound)**.** *In a profile with $d_{\text{top}} \leq \eta$, no leaf outside $\mathscr{D}(A_\eta(u))$ can lie between two leaves of u. Every pair in $\mathscr{P}_\eta$ lies wholly in one block orientation or the other. Consequently,*

$$\mathrm{NROE}(\pi) \geq C_\eta(T,q), \qquad \tau(\pi) \geq C_{w,\eta}(T,q,W), \tag{13}$$

with the second inequality applying to the interval model.

The separation principle excludes interleaving for each eligible pair; the two possible orientations then yield κ and ω. Thus C_η bounds normalized reference-order error, while $C_{w,\eta}$ includes finite-width and boundary costs in coordinate units. Flattening every parent gap of at most 2η and applying Theorems 1 and 2 to the remaining tree gives a second valid certificate. Neither dominates: accumulated gaps can make the chain certificate much stronger (Section 5.2), while flattening also separates pairs of single leaves, which $\mathscr{P}_\eta$ omits. We report the larger of the two.

Scope. The bounds depend only on admissibility (Section 4.1); an allowance changes the displayed hierarchy, never the field or the feature identities. Theorem 1 holds for every map that keeps the merge tree exactly, whatever its width rule; Theorem 2 for exact-topology maps in the interval model $\mathscr{L}$; and Theorem 3 for every map whose merge error stays within η. In an exact-topology map, any order error above ν^* is chosen by the order rule: the tree does not force it. AnchorMTM ranks and admits orders by the exact fixed-order costs of Section 4.4 and uses the bounds as benchmarks.

4.3 Spatial Reference

Cross-frame Leaf Correspondence. We track merge-tree leaf extrema p_i^t using their leaf-arc supports S_i^t. On a shared fixed grid, overlap-based preprocessing [9] counts common spatial samples:

$$O_{ij}^t = |S_i^t \cap S_j^{t+1}|. \tag{14}$$

Hungarian assignment maximizes total intersection count over one-to-one pairs, discarding zero-overlap pairs.

Continuation inherits a Track ID; *birth/death* starts an unmatched current identity or ends an unmatched previous track. Only one-to-one links are retained; identity states do not identify physical split or merge events. For a current leaf i, $\mu_t(i)$ denotes its valid predecessor.

The correspondences are computed once, before any budget sweep, and supply the motion samples for the reference. We distinguish the reference target q_i^t from the display anchor x_i^t. By default, one unit direction $a^* \in \mathbb{R}^d$, $d \in \{2,3\}$, is estimated offline from all matched consecutive-frame extremum displacements:

$$\Delta p_i^t = p_i^t - p_{\mu_t(i)}^{t-1}, \qquad S = \sum_{t,i \text{ matched}} \Delta p_i^t (\Delta p_i^t)^\top, \qquad a^* \in \arg\max_{\|a\|=1} a^\top S a. \tag{15}$$

The leading eigenvector maximizes squared projected motion; it does not consider the tree, and Section 5.6 uses the bounds to compare it with other directions. The same calculation serves 2D and 3D and runs once per sequence. Sign, degenerate and tied cases follow fixed rules given in the supplement. An analyst may supply a physical direction instead; the rest of the construction is unchanged.

For the selected direction, the minimum and span of the projected domain bounding-box corners define a common origin o and extent $L > 0$. The reference is

$$q_i^t = \langle a^*, p_i^t \rangle - o \in [0, L]. \tag{16}$$

Direction, origin and scale stay fixed across frames and budgets. Distinct extrema may share a reference, and anchors approach their references as far as the constraints allow.

Relative geometry uses distances d_{ij}^t between leaf-support centroids. These differ from extremum references because a changing support can move its centroid while the extremum stays fixed.

4.4 Spatially Anchored Layout Construction

Figure 3 shows the four stages. All remaining settings belong to a fixed configuration Θ_0: position slack β_0, the four objective weights, width scale, gap, anchor slack and output resolution. Only ε varies in a budget sweep; Supplement S1 gives the defaults and tolerances.

With features and their reference fixed, construction chooses an order by the lexicographic objective

$$\begin{aligned} \pi_t^{\varepsilon} \in \arg\min_{\pi \in \mathscr{C}_t(\varepsilon)} & \big(\tau_t(\pi), \mathrm{NROE}_t(\pi)\big), \\ \text{s.t.} \quad & \delta_t^*(\pi) \le \eta(\varepsilon). \end{aligned} \tag{17}$$

The finite candidate pool contains the original strict-topology order, midpoint initializations, and orders visited during width-aware refinement of threshold and selective flattenings. The search is heuristic; Section 5.6 compares it with exhaustive enumeration on small trees.

Algorithm 1: Single-Control AnchorMTM

Data: 2D/3D field sequence; $\varepsilon \ge 0$; fixed configuration Θ_0.
Result: Temporal map I^{ε} and error records, or a failure report.
Extract trees, supports and extrema; match supports; estimate one direction and references q_t (once)
$\eta \leftarrow (\varepsilon_0 + \varepsilon)\Delta_F$
for *each time step* t **do**
  $\mathscr{F}_t \leftarrow \{\pi \in \mathrm{GenerateAndRefine}(T_t, q_t, W_t, V_t, \eta) : \delta^*(\pi) \le \eta\}$; report if empty
  Select $\pi_t^{\varepsilon} \in \mathscr{F}_t$ by (τ, NROE); set optimal merge barriers
  Solve the QP for π_t^{ε} with cap $\tau(\pi_t^{\varepsilon}) + \beta_0$; reconstruct the profile; record errors
return I^{ε} *and records*

4.4.1 Topology-Budgeted Leaf Ordering

Exact Mode sets $\varepsilon = 0$, leaving only the numerical allowance $\eta = \varepsilon_0 \Delta_F$; on every planar sequence its rendered merge error is zero. *Budgeted Mode* permits $\varepsilon > 0$. Both modes share the same interval model and placement objective.

Candidate generation includes the exact tree, threshold flattenings, and a conflict-driven selective flattening. The midpoint orders of Theorem 1 initialize these structures, so the exact-tree candidate already attains ν^*. Width-aware block insertion refines the four best initial structures, together with the exact tree, while retaining the point-model budget order as a candidate. Moves improve the lexicographic score consisting of τ and a squared packing-residual term, and must pass $\delta^*(\pi) \le \eta$.

Guards report capacity failures, empty admitted sets and solver failures instead of silently rescaling widths, gaps or canvas; the supplement details them.

4.4.2 Optimal Merge Barriers and Topological Feasibility

For a fixed order, a pair's displayed merge is the largest intervening barrier. Set $C_k = \min_{i \le k < j} V_{\pi(i)\pi(j)}$ and $F_k = \max(V_{\pi(k)\pi(k)}, V_{\pi(k+1)\pi(k+1)})$ for gap k.

Proposition 1 (Optimal barriers)**.** *With fixed extremum values in the continuous non-strict model, the smallest labelled merge error is*

$$\begin{aligned} \delta^*(\pi) = \max \Big\{ & 0, \max_k (F_k - C_k), \\ & \frac{1}{2} \max_{i<j} \Big(V_{\pi(i)\pi(j)} - \max_{i \le k < j} C_k \Big) \Big\}, \end{aligned} \tag{18}$$

and barriers $b_k = C_k + \delta^*(\pi)$ *attain it.*

Error δ requires $F_k \le b_k \le C_k + \delta$. Choosing the largest admissible barriers satisfies every pair's lower merge requirement exactly when $\delta \ge \delta^*$. Thus the order meets the merge-level allowance in this continuous model exactly when $\delta^*(\pi) \le \eta$.

4.4.3 Spatially Anchored Spatiotemporal Placement

Once the order is fixed, the core terms are E_R, mean squared deviation from the fixed reference, and E_M, mean squared error in matched projected motion. The auxiliary term E_G approximates support-centroid distances through ordered anchor separations; E_C regularizes anchor offsets from interval centers. All four terms and their nonnegative weights are retained:

$$\begin{aligned} \min_{x_t, z_t} \quad & \lambda_R E_R + \lambda_G E_G + \lambda_M E_M + \lambda_C E_C \\ \text{s.t.} \quad & (x_t, z_t) \in \mathscr{L}(\pi_t), \\ & \|x_t - q_t\|_\infty \le \tau(\pi_t) + \beta_0. \end{aligned} \tag{19}$$

For each valid match, set $\Delta q_i^t = q_i^t - q_{\mu_t(i)}^{t-1}$ and use the previous display anchor plus this reference displacement as the temporal target:

$$\hat{x}_i^t = x_{\mu_t(i)}^{t-1} + (q_i^t - q_{\mu_t(i)}^{t-1}). \tag{20}$$

Thus E_M follows projected motion and keeps $x - q$ continuous; unmatched leaves, including those of the first frame, have no motion term.

Only ε changes the topology allowance $\eta(\varepsilon)$. The fixed slack $\beta_0 \ge 0$ bounds additional position deviation after ordering. The cap is feasible because $\tau(\pi_t)$ is attained, and the QP keeps the selected order. For a fixed order the problem is convex; frames are solved in temporal order, so the sequence is not optimized jointly.

4.4.4 Profile Reconstruction and Output Diagnostics

Reconstruction inherits the leaf-arc sampling and LCA-path filling of the merge tree map pipeline [9, 22]. Original extrema remain at their anchors; regular samples provide scalar context within intervals and gaps. Optimal-barrier correction then clips each inter-anchor interior to b_k and raises an existing peak if necessary.

A common direction, origin, extent, and pixel conversion are used for all frames. The profiles form the final temporal map,

$$I^{\varepsilon}(t, u) = \psi_t^{\varepsilon}(u), \qquad 0 \le u \le L. \tag{21}$$

Resampling, strict-extremum clearance and raster rounding can change the achieved error, so records report rendered errors separately from δ^*.

5 Experiments

The experiments ask whether the bounds are accurate (Section 5.2), how AnchorMTM compares with existing merge tree maps (Section 5.3), what a merge-level allowance buys (Section 5.4), how the maps read (Section 5.5), and what the choice of reference and the construction cost (Section 5.6).

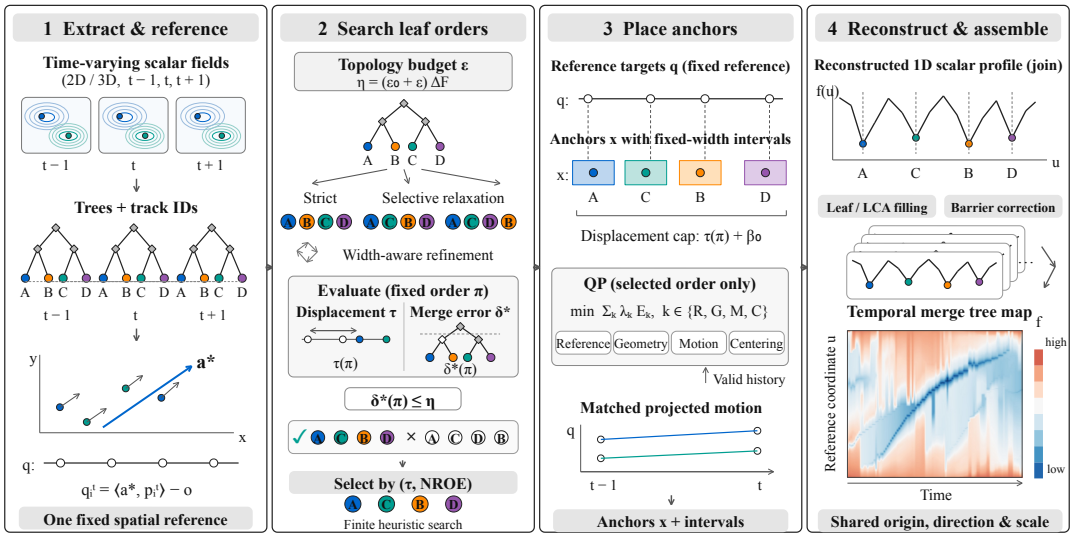


Figure 3: AnchorMTM construction. (1) Extract augmented merge trees, match leaf supports across frames and estimate one fixed spatial reference from matched extremum motion. (2) Generate and refine candidate leaf orders, admit those with $\delta^*(\pi) \le \eta$ and select by (τ, NROE). (3) Place anchors in nonoverlapping intervals with a quadratic program capped at $\tau(\pi) + \beta_0$, using matched history. (4) Reconstruct scalar profiles with optimal-barrier correction and assemble the temporal map on a shared origin, direction and scale. Diagrams are schematic; the map panel is a Storms output at $\varepsilon = 10\%$.

5.1 Experimental Setup

Data. The benchmark comprises seven sequences and 917 frames (Table 1): reanalysis pressure, satellite radiation, fire emission and air quality on planar domains, two volumetric flow simulations, and the synthetic Ring sequence of TMTM [7, 10, 14, 19, 22, 23, 30, 36]. All maps share the processed fields, augmented trees, leaf supports, correspondences and references. References follow Eq. 16 with the estimated motion direction (Eq. 15) on every sequence except Cylinder, where we supply the streamwise axis; on Storms the estimated direction lies 1.1° from west–east. The 744 hourly Storms fields of Figure 1 illustrate the map and enter no benchmark aggregate.

Compared methods. AnchorMTM runs in Exact Mode ($\varepsilon = 0$) and Budgeted Mode ($\varepsilon = 10\%$ unless stated) with its default configuration Θ_0. We compare it with the native outputs of TMTM [22] and ST-MTM [9]. TMTM uses our port of its official implementation, which matches the official core in all 3127 arc intervals and anchors of the planar sequences. ST-MTM has no public code; our implementation of its description orders leaves by optimal leaf ordering [2] and temporal concordance, meets the optimality conditions of its quadratic programs (relative KKT residuals below 10^{-6}) and uses one untuned setting on all sequences (Supplement S10). Each native map is oriented once, from its first frame (Supplement S10). The *projection* places every leaf at its reference and ignores the tree. Native outputs also differ from AnchorMTM in widths and canvas; to compare objectives alone, Figs. 6 and 7 and the Storms case solve two objectives modelled on these methods in our interval model with exact topology and no reference term or cap: temporal alignment (*TMTM-style*) and support-centroid distances with temporal alignment (*ST-MTM-style*); they do not reproduce the original methods. On the planar sequences we also order the extrema along Hilbert and Morton curves. On OLR we also draw a Hovmöller diagram [20] along the same reference, averaging the field over lines perpendicular to it (Supplement S13).

Measures. Position is measured by NROE (Eq. 3), which compares maps with different native scales by their order alone, and, for maps in the interval model, by the pooled mean E_{pool} and the worst E_∞ of $|x_i - q_i|/L$ over all feature–frames, and by NPE, the worst $|x_i - q_i|/L$ of a frame averaged over frames. Topology is measured by NMLE, the rendered $d_{\mathrm{top}}/\Delta_F$ (Eq. 5). Temporal behaviour is measured by the motion error (ME), the fraction of clear approach or separation events, i.e., changes of the 2D or 3D distance between two matched extrema over two steps, that a map shows in the wrong direction or not at all; by TD, the summed map displacement of matched features [9]; and by the layout-induced jump $|\Delta x_i - \Delta q_i|/L$ of matched features per step. To credit the objective of ST-MTM, we also report scale-normalized stress (SNS) and trustworthiness (TW, $k = 1$) of anchor distances against support-centroid distances with its protocol [9]. TMTM and ST-MTM keep the merge tree exactly, so we measure them against the strict bound $\nu^*(T, q)$. Exact Mode keeps the numerical allowance, so we measure it against $c_0 = C_\eta$ at $\eta = \varepsilon_0 \Delta_F$ (Theorem 3), which its order search attains on all 917 frames. The two bounds differ only on Cylinder, where near-tied saddles lower the bound from 12.24% to 8.01% at a merge error below $10^{-7}\Delta_F$ (Supplement S16).

Table 1: Benchmark sequences. Leaves: retained extrema per frame.

Sequence	Scalar field	Frames	Leaves	Tree
Storms	ERA5 mean sea-level pressure, 2D	118	1–8	join
Wildfires	log(1 + FRP), 2D	30	6–11	split
OLR	outgoing longwave radiation, 2D	60	2–8	join
$PM_{2.5}$	$PM_{2.5}$ concentration, 2D	366	1–7	split
Ring	synthetic ring intensity, 2D	40	1–7	split
Tangaroa	velocity magnitude, 3D	201	18–93	join
Cylinder	Q criterion, 3D	102	4–75	split

5.2 The Bounds Are Attained in Controlled Trees and Informative on Real Ones

Each bound isolates one cause of displacement. To separate the causes, we fix two sibling pairs $X = \{A, B\}$ and $Y = \{C, D\}$ that merge at 0.4 below a root at 1, with leaf values 0 and references $(0.2, 0.5, 0.5 - d, 0.8 - d)$ on a unit canvas, and vary the signed overlap d from -0.12 to 0.30. Across 43 overlaps and

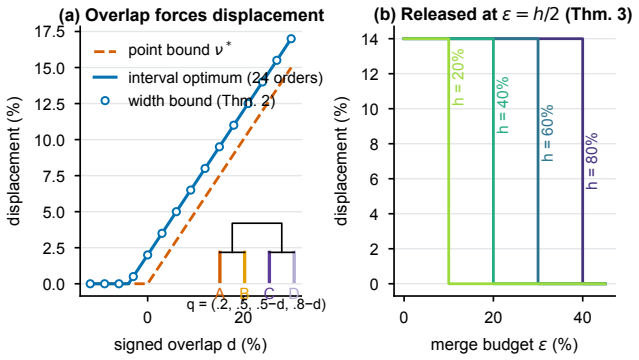


Figure 4: Controlled four-leaf trees (inset: sibling pairs and references). (a) Increasing sibling-range overlap d forces displacement; the width bound of Theorem 2 (circles) equals the optimum over all 24 orders. (b) At $d = 24\%$, the 14% displacement disappears exactly when the merge budget reaches half the parent gap h, for all four gaps, as Theorem 3 predicts. Percentages refer to a unit canvas or a unit scalar range.

four widths, we compare the bounds with all 24 leaf orders, each placed by an independent minimax LP. Topology alone forces exactly $\nu^* = \max(0, d)/2$. With widths $W = 0.06$, gap 0.01 and $\rho = 0.5$, the width bound, our search and the exhaustive optimum coincide at $\tau = \max(0, (d + 0.04)/2)$ (Fig. 4a; Fig. 2c): 12% and 14% at $d = 0.24$. A merge allowance removes the conflict exactly where Theorem 3 admits interleaving, at $2\varepsilon = h$ for all four parent gaps tested (Fig. 4b).

On real trees, the bounds certify most selected orders as optimal. In Exact Mode, the interval cost τ of the selected order equals the width bound on 606 of the 614 planar frames, so on these frames no exact-topology order admits a smaller worst displacement; at $\varepsilon = 10\%$ the count is 596. The count stays between 578 and 608 when the width scale is halved or doubled, the gap removed or doubled, or the anchor slack set to 0 or 1. Volumetric trees are harder: the bound is met on 81 of 102 Cylinder frames and on 27 of 201 Tangaroa frames (Supplement S10). Accumulated ancestor gaps keep the bounded-topology bound informative on these large trees: on Tangaroa at $\varepsilon = 5\%$, flattening every gap below 2η certifies only 0.38% on average, whereas Theorem 3 with widths certifies 7.40% against an attained 8.10%. The remaining gap does not tell slack in the bound from a suboptimal search.

5.3 Comparison with Existing Merge Tree Maps

Most of the order error of TMTM and ST-MTM is not forced by the tree. Both methods keep the merge tree, so Theorem 1 splits their order error into the part ν^* that the hierarchy forces and an excess chosen by their order rule, which the midpoint order of the same tree removes. The excess is 54–95% of TMTM's order error and 31–95% of ST-MTM's (Fig. 5a); the smallest shares are on the volumetric sequences, where the hierarchy forces more. The shares do not hinge on ST-MTM's setting or on the orientation of the native maps: over every value of the reordering threshold, its only parameter that changes the order, and in both orientations, they stay at 73–97% on Storms, OLR and $PM_{2.5}$, 53–82% on Wildfires and 37–75% on Ring (Supplement S10). On Storms, TMTM's 19.2% contains 3.8% that the hierarchy forces, and 12.5% of ST-MTM's 16.3% is not forced. Exact Mode matches the bound to within 0.03 percentage points on every planar sequence with zero merge error, as its midpoint initialization leads one to expect, and attains c_0 on Cylinder. Space-filling curves do not close the gap: Hilbert and Morton orders of the extrema, averaged over eight orientations, have NROE of 8.3–23.4% on the planar sequences and, even with optimal barriers, change merge levels by up to 19–39% of the scalar range (Supplement S10).

The anchored axis carries motion. The motion term rewards displayed motion that follows projected motion. Exact Mode shows fewer approach and separation events wrongly than TMTM and ST-MTM on every sequence: by wide margins on Storms (24% against 41–47%), OLR (5.4% against 21–45%), Tangaroa and Cylinder, and by one to four events on Wildfires, $PM_{2.5}$ and Ring. The projection, which follows the reference order exactly, is lower still on five sequences, but only by breaking the tree: its merge errors reach 59% of the scalar range (Fig. 5b), and on Tangaroa no frame keeps its tree. Budgeted Mode closes part of this gap within its allowance (Section 5.4), and no rendered AnchorMTM map exceeds its declared allowance beyond numerical tolerance.

Anchoring costs pairwise-distance fidelity. One reference direction cannot preserve distances in two or three dimensions. ST-MTM, whose objective is pairwise distance, has the best TW on all seven sequences and the best SNS on five; the space-filling curves are worse than ST-MTM in both on every planar sequence. Where widths dominate, the trade-off can reverse: on Cylinder, Budgeted Mode reaches an SNS of 0.041 against ST-MTM's 0.135.

5.4 One Control Traces the Position–Topology Trade-off

One control spans the trade-off from exact topology to the reference order, and the depth of the conflicting merges decides how fast position returns. Figure 6 traces each sequence from Exact Mode to the projection; every point respects its allowance. Theorem 3 lifts the restriction on two subtrees once 2η reaches their merge gap (Fig. 4b). Where conflicting subtrees merge at shallow levels, small budgets suffice: on Cylinder, $\varepsilon = 1\%$ lowers the worst displacement from 27.00% to 5.27%, below the projection's 5.73% because sorting by reference ignores widths; on Wildfires, 1% lowers E_∞ from 20.48% to 8.71%; on $PM_{2.5}$, 5% already reaches the projection. Where they merge deep, position is expensive: on Storms the worst displacement stays above 18% up to $\varepsilon = 20\%$, and on OLR one frame keeps $E_\infty = 17.14\%$ up to 10%. Budgets are spent where they are needed: on Storms at 10%, the mean merge error is 0.90% while the worst frame uses 8.51%. Larger budgets also redistribute displacement, so some features move away from their references while the mean falls (Supplement S15). The TMTM-style and ST-MTM-style objectives, which have no reference term, lie far above every curve (grey boxes; Supplement S11).

5.5 Reading the Maps

OLR: features on their references, with their values and nesting. Figure 7 shows the same 60 days in a Hovmöller diagram and in four merge tree maps. Among the merge tree maps, only the anchored ones keep the low-radiation basins on their references: the TMTM-style and ST-MTM-style objectives leave mean displacements of 24.4% and 42.7% of the axis, Exact Mode 1.0%. The Hovmöller diagram places every sample exactly but averages across the reference. Of the 264 minima, its profile shows a separate dip within 2% of the reference for 59%, and these dips are on average $32\,\mathrm{W\,m^{-2}}$ shallower than the minima; minima at similar reference positions but different latitudes merge into one dip. By the same test, Exact Mode has a dip for 94% of the minima, and 93% are shown at their exact values; the others are displaced farther or swapped with a neighbouring minimum. Their nesting is kept. The Hovmöller diagram is smoother in time: a minimum appears or vanishes between every pair of consecutive days, and a map that keeps each day's tree changes with it (mean column change 9.0% of the scalar range, against 4.1%). The two displays answer different questions and can be read side by side.

Storms: an exact map along the west–east axis. On the 118-frame Storms benchmark, the minima are low-pressure systems crossing the North Atlantic, and the reference runs west–east. Mea-

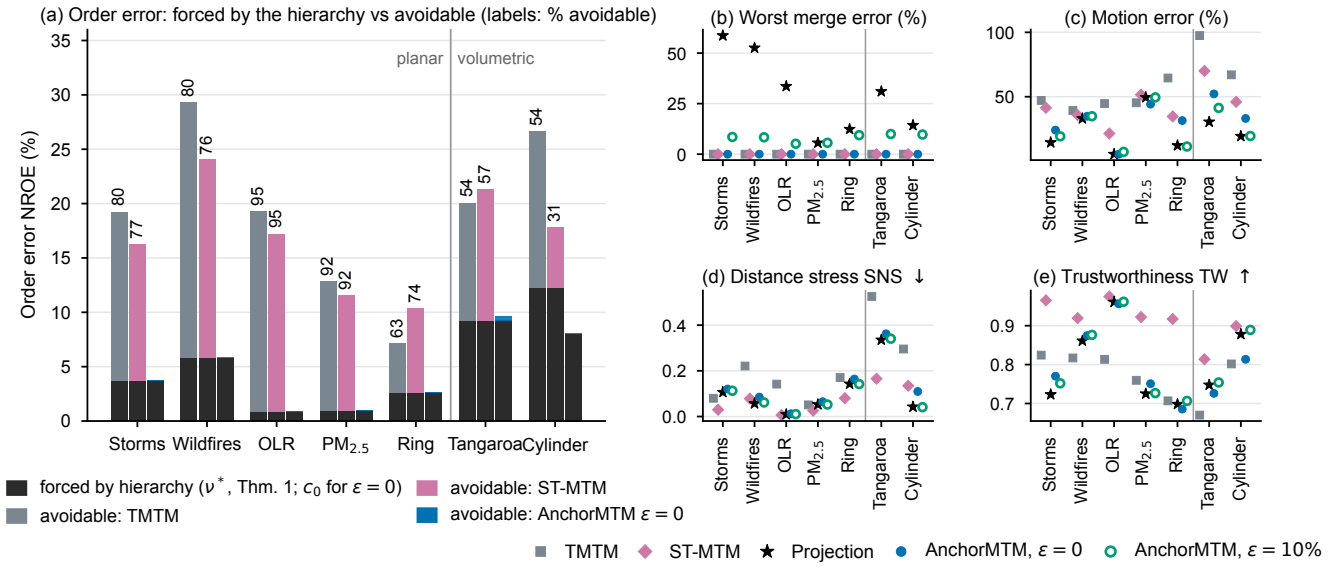


Figure 5: Native outputs of TMTM and ST-MTM, the projection and AnchorMTM on seven sequences (left of the grey lines: planar; right: volumetric). (a) Order error NROE split by Theorem 1 into the part forced by the hierarchy (black: ν^* for TMTM and ST-MTM, c_0 for Exact Mode; they differ only on Cylinder, Section 5.1) and the avoidable excess of each map; labels give the avoidable share in percent. (b) Worst rendered merge error: exact-topology maps have none, the projection up to 59%. (c) Motion error. (d, e) The distance measures of ST-MTM. All values are listed in Supplement S10.

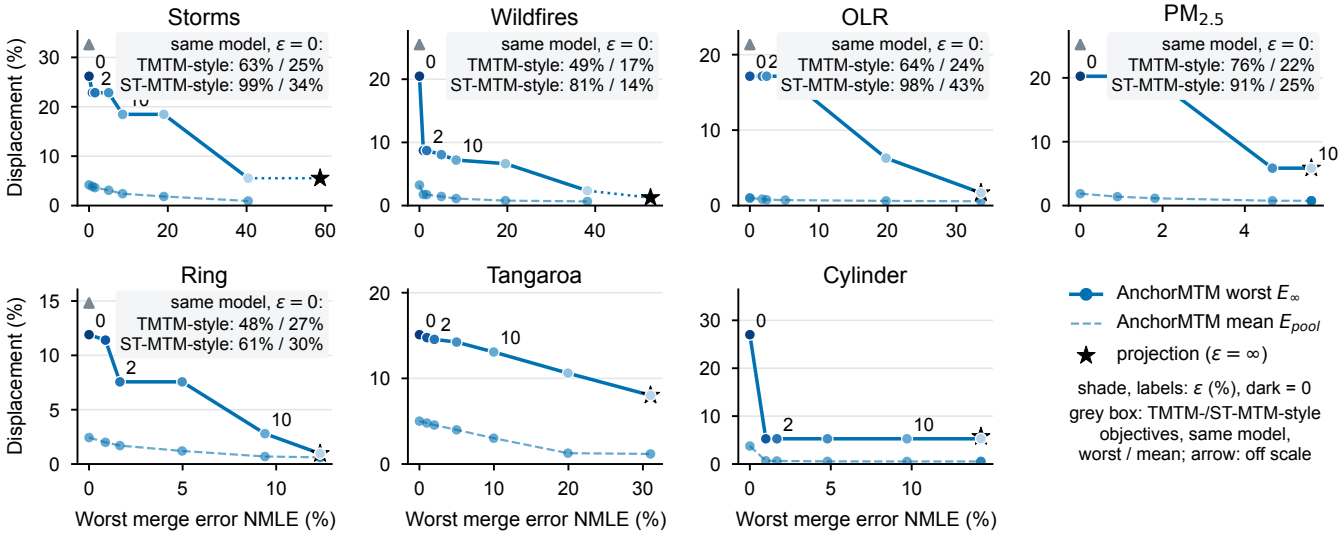


Figure 6: Position–topology trade-off traced by the single control ε. Each point is one map of all frames at budget ε (shade and labels, in percent), placed by its rendered worst merge error and its worst (E_∞, solid) and mean (E_{pool}, dashed) displacement. Stars: projection ($\varepsilon = \infty$). Grey boxes: worst and mean displacement of the TMTM-style and ST-MTM-style objectives (Section 5.1), above the plotted range (arrows).

sured by NPE, the TMTM-style and ST-MTM-style objectives place features 37.3% and 50.7% of the canvas from their references within the same layout model, and Exact Mode 5.1% (median per frame 0.8%, peaks of 26.2% in a few frames). A 10% budget lowers NPE to 2.7%; it is spent in 31 of 118 frames, at most 8.51% and 0.90% on average, and the 89 barriers whose merge level changed by more than 0.1% of the scalar range are marked on the map (Supplement S12). Figure 1 applies the construction at $\varepsilon = 5\%$ to 744 hourly fields of December 1999 with a fixed projected direction. Its snapshots follow one matched minimum, a strong storm over Scandinavia, from 16 to 19 December, whose displayed positions form one continuous track through the outlined window. The track is a correspondence of leaf features; it does not by itself identify a physical storm. Supplement S15 follows one tracked feature per sequence through six budgets.

5.6 Reference Choice, Ordering and Cost

Because the bounds need no layout, they inform the choice of reference before a map is drawn. In 3D, a projection discards two coordinates, and the unavoidable order error depends on the direction. We evaluated the bound c_0 for 400 sampled directions on both volumetric sequences, with references built as in Eq. 16 (Supplement

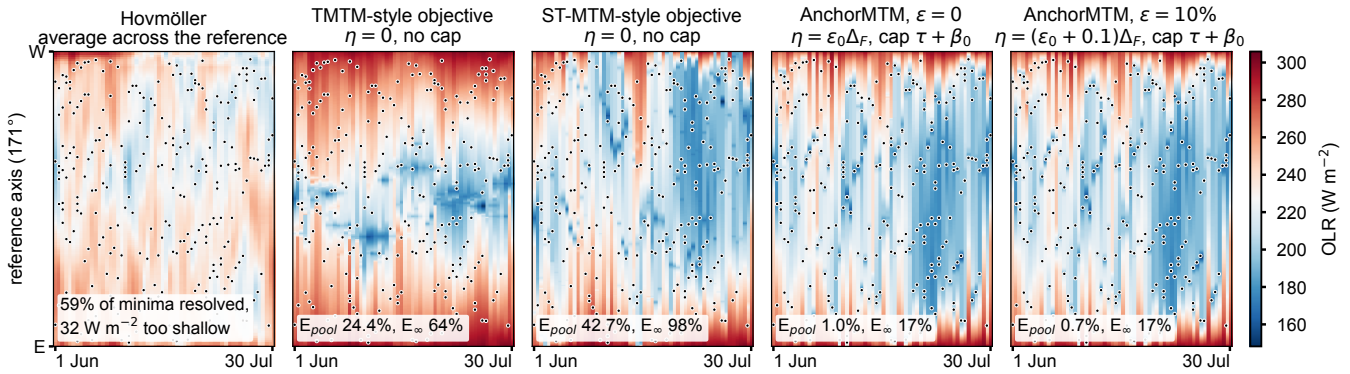


Figure 7: Sixty days of OLR (June–July 2024, tropical Indian Ocean and western Pacific) on one reference axis, the estimated direction 171°, running from east to west. Left: Hovmöller diagram along the reference. Right: four merge tree maps, labelled by their allowance and cap. Black dots: reference positions of the minima. All panels share one colour scale; labels give the share of minima resolved by the Hovmöller profile within 2% of their reference and the mean depth error of those dips, or the displacement of the merge tree maps.

S14). Over these samples its mean ranges from 4.70% to 10.14% on Tangaroa and from 5.49% to 28.15% on Cylinder. On Tangaroa, the estimated motion direction is among the most conflicting: only 4 of the 400 sampled directions have a higher bound, and the x axis would lower it from 9.24% to 6.09%. On Cylinder, the streamwise axis lies near the median (8.01%). In every key direction, Exact Mode stays within 0.6 percentage points of the bound, except on the Tangaroa z axis (2.6 points), where large widths dominate, while TMTM and ST-MTM stay far above it.

Width-aware ordering. An order that is optimal for points need not be optimal for intervals when features are large. On Tangaroa, ordering by NROE alone, i.e., the point-model budget order, leaves a worst displacement of 20.69% at $\varepsilon = 0$. Ranking candidates by the exact interval cost τ, as Algorithm 1 does, lowers it to 15.10% at $\varepsilon = 0$, at a slightly higher NROE (9.67% against 9.24%), and to 14.55% at $\varepsilon = 2\%$, where the worst frame's order meets the width bound. On Cylinder, $\varepsilon \geq 2\%$ lowers E_∞ from 5.73% with the point-model order to 5.27%, again at the bound.

Search coverage and runtime. On all 637 frames with at most eight leaves, the point-model order search attains the exhaustive NROE optimum at nine budgets from 0 to 100%. The bounds cost $O(n \log n)$ to $O(n^2 \log n)$ and δ^* costs $O(n^3)$ per order. On an Apple M2 Pro (one Python process; NumPy 2.0, SciPy 1.13), the bounds take at most 0.6 ms per planar frame and 5.5 ms per volumetric frame (up to 93 leaves), and all 917 frames take 94 s at $\varepsilon = 0$ and at most 137 s with a budget, most of it in the capped QP.

6 Discussion and Limitations

What the bounds are for. The bounds need neither a layout nor a solver. Before a map is drawn, they compare candidate references in milliseconds (Section 5.6); afterwards, they separate the displacement that the hierarchy forces from the part another layout could remove, for any map that meets their constraints. Order error above ν^* is always removable without changing the tree, since midpoint ordering attains ν^*; the other two bounds need not be attained, so the excess above them overstates what a better layout could remove.

Stability and motion. Temporal stability measures reward maps that do not move. On Storms, the projection, which follows the references, accumulates a TD of 23.1 canvas units, while TMTM and ST-MTM move their features by only 10.1 and 7.2 and show motion most often wrongly (Section 5.3). AnchorMTM's movement mostly follows the reference: on OLR, the layout-induced jump averages 1.0% of the axis per step against an anchor motion of 5.9%. On volumetric data, layout jumps are as large as the anchor motion (2.8% per step on Tangaroa), a cost that a temporal term in the order selection could reduce.

What a budget changes. A budget never removes a feature, changes an extremum value or moves the data; it changes merge levels by at most $\eta = (\varepsilon_0 + \varepsilon)\Delta_F$, and every rendered map stayed within this allowance. Where the labels cover all leaves, as we verified on 200 rendered columns of Ring and OLR, the displayed tree and its persistence diagram stay within the measured merge error of the source in interleaving and bottleneck distance. Persistence simplification at threshold 2η would instead delete the features below it [12, 33]. A split subtree no longer forms one contiguous band, which is why such barriers are marked.

Using AnchorMTM. We suggest three steps. Compare candidate references by ν^*, which takes milliseconds per frame. Start in Exact Mode, whose merge tree is exact and whose order is optimal on almost every planar frame. Then sweep ε, take the smallest budget at which the trade-off curve drops, and inspect the marked barriers before accepting it. For reading propagation speed, a Hovmöller diagram along the same reference complements the map.

Limitations. One direction cannot encode the other coordinates, and the estimated direction, which follows the dominant motion, need not be the least conflicting one. It is estimated offline from all matches, so online use needs a supplied direction. The construction inherits the correspondence: missed or spurious matches change the estimated direction and the motion term. The order search is a finite heuristic, checked exhaustively only on small trees (Section 5.6). Certification depends little on Θ_0, but displacement scales with the chosen widths (Supplement S10). Temporal behaviour and relative geometry are measured but not certified. Finally, the motion error is a geometric proxy that assumes known identities; whether analysts read position and nesting together, and whether disclosure marks prevent misreading [21, 25], requires a user study.

7 Conclusion

AnchorMTM gives temporal merge tree maps a spatial axis and states its price. Three bounds give the displacement that exact topology, finite widths and a bounded merge error force on any layout. On seven sequences, 31–95% of the order error of TMTM and ST-MTM is not forced by the tree, and the order of Exact Mode meets the width-aware bound on 606 of 614 planar frames. Next steps are scalable order search, references beyond one direction, temporal certificates and a user study.

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